Data Analysis

Syllabus

Yu-You Liou

Fall 2026

Lecturer: Yu-You Liou	**Contact**: d10627008@ntu.edu.tw
Course Code: EIB-301-01-A1	**Website**: https://yyliou.github.io/da
Meeting Time: Tuesday, Periods 8–9 (15:10–17:00)	**Location**: A207

Course Objectives

1. To enable students to understand the basic logic, applications, and simple techniques of data analysis using R.
2. To develop students' ability to build clear and informative data visualizations grounded in the grammar of graphics.
3. To communicate data-driven findings clearly to both technical and non-technical audiences.
4. To integrate generative AI to enhance students' workflows in programming and compilation tasks.

Textbooks

1. Wickham, H., Navarro, D., & Pedersen, T. L. *ggplot2: Elegant Graphics for Data Analysis* (3rd ed.). (Main) https://ggplot2-book.org
2. Adler, J. *R in a Nutshell: A Desktop Quick Reference* (2nd ed.). O'Reilly Media.

Course Structure

1. **Lecture** (First hour): Foundational knowledge and concepts introduced each session.
2. **Lab** (Second hour): Hands-on in-class exercises using R to apply the concepts learned during the lecture.

Assessment Criteria

1. **In-Class Exercises** (36%): 12 exercises × 3% each, completed during the lab hour and uploaded to the course pCloud folder (see the deadline under Regulations).
2. **Mid-Term Exam** (32%) and **Final Exam** (32%). Both examinations are conducted online (see the exam regulations on the course website).

Regulations

1. **Attendance**

Per Shih Chien University regulations, students who are absent for more than one-third of class sessions will receive a final grade of zero. Make-up attendance will not be permitted.

2. **Examination**

Students who do not show up for their scheduled examination will receive a grade of zero for that component.

3. **In-class exercise deadline**

The deadline for each in-class exercise is 23:59 (UTC+8) on the day before the following week's class. Late submissions will not be accepted.

4. **Academic Integrity**

All submitted work must be the student's own. Copying from other students is strictly prohibited. Every submission carries a unique identifier generated by the official template, and identical identifiers across submissions are treated as evidence of copying; any work found to be copied will receive a grade of zero for all students involved.

5. **AI Policy**

Students may use AI tools (e.g., ChatGPT, Gemini, GitHub Copilot) as learning aids, and doing so is not itself a violation. What is required is that you can explain and defend any code or analysis you submit. Submitting AI-generated work you do not understand constitutes academic dishonesty.

Required Software

All software is free and open-source: **R** at https://cran.r-project.org and **RStudio Desktop** at https://posit.co/download/rstudio-desktop. The R packages needed for the course, including the ones every submission depends on, install with a single command: `install.packages(c("tidyverse", "patchwork", "uuid", "rmarkdown", "knitr", "tinytex"))`, followed by `tinytex::install_tinytex()` for PDF output. Step-by-step instructions are on the course R page.

Course Schedule

1. Weeks 1–2: Part I — Introduction to R and first steps with `ggplot2`.
2. Weeks 3–7: Part II — Layers (individual geoms, collective geoms, statistical summaries, maps, networks, annotations, arranging plots).
3. Weeks 8–9: **Mid-Term Examination** (online; the exact date will be announced on the course website).
4. Weeks 10–11: Part III — Scales (position scales, colour scales, other aesthetics).
5. Weeks 12–14: Part IV — The grammar; advanced topics.
6. Weeks 15–16: **Final Examination** (online; the exact date will be announced on the course website).
7. Weeks 17–18: Flexible learning weeks.

Syllabus is subject to change. Any updates will be announced on the course website.